

Practical Cybersecurity & Ethical Hacking Foundations

Aligned with CEH v13 | CompTIA Security+ | NIST Cybersecurity Framework

Duration: 40 Hours | **Format:** Practical + Instructor-Led | **Level:** Beginner to Job-Ready

A practical, hands-on introduction to cybersecurity for the modern workplace.

Participants gain real experience in **threat detection, vulnerability analysis, system hardening, cloud protection, and incident response**, aligned with **ISO 27001, NIST CSF**, and **SOC 2** standards recognized worldwide.

1. Cybersecurity Fundamentals

Learn the basics of cyber defence — from how attacks happen to how organizations protect their data.

Students explore ethical hacking principles, threat types, and compliance awareness through local and global case examples.

Keywords: Threat landscape, CIA triad, risk posture, cyber hygiene.

2. Network Security & Monitoring

Understand how data travels through a network and how to detect suspicious activity.

Hands-on practice analyzing traffic and identifying intrusion patterns.

Keywords: Wireshark, tcpdump, network forensics, firewall tuning.

3. Information Gathering & Reconnaissance

Learn to perform responsible information gathering using publicly available data.

Students apply OSINT techniques and identify exposed digital assets safely.

Keywords: Footprinting, WHOIS, DNS lookup, OSINT automation.

4. Vulnerability Assessment & Risk Prioritization

Find and evaluate security weaknesses using real tools.

Understand how to interpret scan results, rate risks, and communicate fixes clearly.

Keywords: Nmap, Nessus, CVE mapping, remediation planning.

5. Operating System Security & Privilege Management

Secure Windows and Linux systems through access control and auditing. Apply least-privilege concepts, review logs, and enforce user policies.

Keywords: Privilege audit, log review, hardening, access management.

6. Web & Application Security Essentials

Understand common website and API vulnerabilities and how to prevent them. Students test safely within a lab setup and learn secure coding principles.

Keywords: OWASP Top 10, XSS, SQLi, input validation, secure coding.

7. System Hardening & Security Monitoring

Reduce attack surfaces and detect misuse through continuous monitoring. Students implement configuration baselines and basic alerting techniques.

Keywords: CIS Benchmarks, patch management, log correlation, SIEM basics.

8. Cloud Security Fundamentals

Explore how to secure accounts and data in cloud environments like AWS and Azure. Students learn access control, encryption, and configuration best practices.

Keywords: IAM, cloud misconfiguration, data encryption, cloud audit.

9. Incident Response & Reporting

Understand how security teams detect and respond to real-world cyber incidents. Students follow the NIST process: Detect → Contain → Recover → Review.

Keywords: IOC identification, containment, forensic triage, NIST 800-61.

10. Security Assessment & Career Path Planning

Conduct a complete cybersecurity assessment and report findings professionally. Map your next certifications and career direction — SOC Analyst, Security Engineer, or CEH.

Keywords: Recon-to-report, vulnerability lifecycle, SOC readiness, job roadmap.

11. Bug Bounty & Responsible Disclosure

Learn how global companies reward ethical hackers for finding vulnerabilities. Understand disclosure policies, report writing, and communication ethics.

Keywords: Bug bounty, disclosure process, report validation, platform standards.

PROGRAM OUTCOMES

- Perform professional, ethical security testing aligned with global standards.
- Detect and mitigate vulnerabilities using real cybersecurity tools.
- Apply ISO 27001, NIST CSF, and SOC 2 controls in practical situations.
- Build a portfolio that demonstrates **hands-on, job-ready cybersecurity capability**.